

Prediction or Revaluation? Phase- and Space-Aware VAEP for Women’s Football

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Abstract. Event-based action valuation models such as VAEP value on-ball actions from their type, outcome, location and recent context, but do not encode the off-ball spatial configuration in which the action is performed — a limitation that bites hardest in women’s football, where public spatial data remains scarce. We study whether two interpretable context layers, rule-based tactical phase labels and geometric features from StatsBomb 360 freeze frames, affect VAEP-style valuation in a men-to-women source-target setting.

We compare four variants in a 2×2 design over the two layers on a held-out women’s tournament, with paired match-level bootstrap intervals and across ten training seeds. Phase information improves estimation of conceding risk by +0.006 ROC-AUC averaged over seeds — the only difference in the study separated from zero, and four times smaller than a single run suggests. Spatial features do not improve aggregate discrimination and degrade it when combined with phase, the layers acting as substitutes rather than complements; they nonetheless reorder the player ranking from $\rho = 0.98$ to $\rho = 0.82$, and a stratified analysis does not identify the situations responsible. We characterise the redistribution without treating it as a correction. Source-domain similarity proves governed more by competition type than by gender, and a model built on spatial features predicted from event data matches one built on observed freeze frames.

Keywords: action valuation · VAEP · women’s football · StatsBomb 360 · spatial features · transfer learning · football analytics

1 Introduction

Football action valuation assigns credit to individual on-ball actions by estimating how much they change a team’s probability of scoring and conceding. VAEP is one of the most widely used frameworks for this task because it values all on-ball actions, not just shots [4]. It is, however, primarily event-based: it observes action type, result, location, body part, time, score state and a short history of previous actions, but not the spatial configuration of off-ball players around

the action. This creates two distinct limitations, which motivate two different context layers.

The first is spatial. A pass or take-on performed under pressure, through a compact defensive block, is not equivalent to the same action performed in open space, yet the two are indistinguishable in the event description.

The second concerns the possession regime rather than the local geometry. Canonical VAEP conditions on a fixed-length history of three preceding actions, which represents the recent past but not whether a possession is settled or has just changed hands. Two own-third passes can agree in type, coordinates, score state and all three preceding actions, yet carry very different risk depending on whether they follow a controlled build-up or a turnover. Because phase governs how severely a turnover is punished rather than how likely a goal is at the other end, the omission should bear primarily on the conceding side — a prediction we test in Section 4.2.

Both limitations bite hardest in women’s football, where event data is increasingly available but public spatial data is far more limited than in men’s football. StatsBomb 360 freeze frames offer an intermediate form of spatial context, recording the players visible at each event but not their trajectories [11], which suits compact interpretable features describing local pressure, support, density and zone occupation.

The paper makes three contributions: a compact interpretable extension of VAEP combining phase annotation with freeze-frame spatial features, evaluated in a full 2×2 design that separates the main effects of the two layers from their interaction; a comparison of women-only training, zero-shot transfer, fine-tuning and pooled source-target training, using size-matched comparisons and learning curves to distinguish the value of additional data from that of cross-domain data; and a separation of predictive performance from valuation behaviour, with every comparison carrying paired bootstrap intervals and averaged over ten seeds.

2 Related Work

2.1 Action Valuation in Football

Traditional player evaluation based on goals, assists or expected goals captures only a small fraction of the actions that shape possession. Possession-value and action-value models instead represent football as a sequence of game states and estimate how actions change the value of the resulting state [4, 13].

The most directly related framework is VAEP [4], which represents matches as sequences of standardised on-ball actions and estimates, for each game state, the probability of scoring and of conceding within a fixed action horizon. This makes it possible to value all on-ball actions and to separate offensive from defensive contributions. A parallel line of work values specific action types, most directly the pass-valuation framework of Bransen et al. [2], developed alongside rather than after VAEP; later work extended action valuation to further learning paradigms [13]. All remain grounded in event streams and do not observe the off-ball configuration around the action.

2.2 Spatial Context and Freeze-Frame Data

Spatial context is central to football analytics: dominant-region, pitch-control and expected-possession-value models use player locations to evaluate space occupation, receiving opportunities and the risk or reward of actions [19, 17, 15, 6, 8], and related work learns pitch-level probability surfaces for passing options and action outcomes [7].

The most closely related freeze-frame valuation model is the VAEP360 variant used in Un-XPass [16], from which we differ on three axes: it evaluates passing decisions and creativity, whereas we value the complete on-ball action stream; it consumes the freeze frame through learned surface-based representations, whereas we use compact handcrafted features chosen for interpretability under partial visibility; and our small, distributionally distinct target domain motivates a low-capacity encoding. The routes are complementary — the phase layer is a property of the game state, not of the spatial encoder, and could be supplied to VAEP360 unchanged — and a direct empirical comparison is left to future work.

Most such models assume continuous tracking data, which is rarely public and especially limited in women’s football. Freeze frames instead enable lightweight features such as nearest-opponent distance, defensive density and opponents between ball and goal. We follow this interpretable route but integrate the features into VAEP-style action valuation rather than pass-choice evaluation.

2.3 Source-Target Learning for Data-Scarce Sports Analytics

Transfer learning uses information from a source domain to improve learning in a related target domain [14], and domain adaptation theory relates target-domain error to source-domain error and to the divergence between domains [1]. Training a spatially aware valuation model only on a small women’s 360 corpus risks overfitting, while training only on men’s football risks domain shift. We therefore treat men’s 360 football as a data-rich source domain and women’s football as a data-scarce target domain. To our knowledge, the combination of VAEP-style action valuation, freeze-frame features and men-to-women source-target learning remains underexplored.

3 Data, Models and Training

3.1 Data and Splits

The dataset is Hudl StatsBomb’s public open-data release [18]. Event data describes each on-ball action with type, result, body part, location and timestamp; StatsBomb 360 augments these with a freeze frame giving the pitch positions of every player visible in the broadcast view at the instant of the event, labelled as actor, teammate or opponent, with a polygon delimiting the captured region [11]. Positions are sampled only at event moments and only within the camera’s field

of view, and the features below are designed to be robust to this partial visibility. Because coverage is itself related to pitch position, we additionally record visible-area size, the number of visible players and the actor’s distance to the polygon boundary, and use them in Section 4.3.

3.2 Train, Adaptation, and Evaluation Splits

We organise the StatsBomb data into four corpora with distinct experimental roles (Table 1). Men’s 360 data forms the source domain, women’s 360 data provides target-domain adaptation data, and UEFA Women’s Euro 2025 is reserved as a held-out evaluation tournament. A separate women’s event-only corpus is included only to show the asymmetry between the broad availability of event data and the much narrower availability of freeze-frame spatial data.

Table 1. StatsBomb corpora and experimental roles; the evaluation corpus is held out.

Corpus	Domain Role		Matches	Events	360 frames
Men’s 360 source	Men	Source training	200	752,967	652,881
Women’s 360 adaptation	Women	Target training/calibration	95	331,303	284,305
Women’s 360 evaluation	Women	Held-out test	31	105,611	90,722
Women’s event-only context	Women	Descriptive context only	414	1,386,529	—

The men’s 360 source corpus includes FIFA World Cup 2022, UEFA Euro 2020, UEFA Euro 2024 and Bundesliga 2023/24. The three international tournaments match the target domain’s competition type; Bundesliga was included because it is the largest men’s 360 corpus in the public release — a data-availability decision rather than a claim that club football is the more similar source. Since competition type may matter as much as gender, we treat the composition as a variable: Section 4.1 reports transfer from the international and club subsets separately, with domain-divergence estimates. The women’s 360 adaptation corpus is FIFA Women’s World Cup 2023 and UEFA Women’s Euro 2022; UEFA Women’s Euro 2025 is the held-out evaluation tournament. The event-only corpus covers FIFA Women’s World Cup 2019, NWSL 2018 and three FA Women’s Super League seasons.

The corpus counts in Table 1 refer to raw StatsBomb events, whereas the evaluation counts reported below refer to the SPADL-converted action stream after filtering to VAEP-eligible actions.

3.3 Model Variants

Event-VAEP. Following Decroos et al. [4], Event-VAEP (E-VAEP) estimates two probabilities for each game state s_i reached after action a_i : the probability that the acting team scores within the next $k = 10$ actions, $P_{\text{score}}(s_i)$, and the probability that it concedes within the same window, $P_{\text{concede}}(s_i)$. The value of action a_i is the change these probabilities undergo across it:

$$V(a_i) = [P_{\text{score}}(s_i) - P_{\text{score}}(s_{i-1})] - [P_{\text{concede}}(s_i) - P_{\text{concede}}(s_{i-1})].$$

Each probability is estimated by a separate binary classifier on the SPADL-converted action stream. E-VAEP uses the canonical VAEP feature set — action type, result, body part, start and end coordinates, distance and angle to goal, elapsed time and time-delta, score state, and the same fields for the three preceding actions — and carries no information about any player other than the actor.

Phase-VAEP. Phase-VAEP (P-VAEP) extends E-VAEP with a six-class tactical phase label assigned by explicit rules, for interpretability. A zone default uses the SPADL coordinates: attacking third is *final-third creation*, own third *build-up*, otherwise *progression*. Three rules override it in increasing priority: a defensive action in the own third immediately after a possession change is a *defensive transition*; an action within five seconds of a possession begun by a recovery or counter-attack is a *transition attack*; any restart, or the first action of a possession beginning from one, is a *set piece*. The one-hot label enters the feature set with its interactions with action type, the 21 SPADL types first being grouped into seven families.

Phase-Space VAEP and Event-Space VAEP. Phase-Space VAEP (PS-VAEP) extends P-VAEP with seventeen geometric features computed from each freeze frame, with coordinates normalised so that the acting team attacks left to right on a 105×68 m pitch. They capture four aspects of the visible context — pressure and support, defensive density, defensive structure, and zone context — and use only the players visible in the frame; full definitions and supplementary results are available in the accompanying repository.¹ Event-Space VAEP (ES-VAEP) adds the same features to E-VAEP without the phase layer, completing a 2×2 design. This is what allows each main effect to be estimated separately from the interaction: the phase effect is identified by both (P – E) and (PS – ES), the space effect by both (ES – E) and (PS – P), and the interaction by their difference.

3.4 Training and Evaluation Protocol

All variants share the same classifier. We use LightGBM [12] because the feature set is tabular, low-dimensional and mixes categorical with continuous variables, because both positive classes are rare and the target corpus small, and because gradient-boosted trees are the standard learner in the VAEP literature and in the reference implementation. Hyperparameters (63 leaves, learning rate 0.05, up to 1500 rounds with early stopping after 50) are taken from that configuration and held fixed across variants and strategies rather than tuned per variant, so that differences are attributable to the feature set alone; Section 4.3 reports the sensitivity to this choice. The two heads are trained as independent binary classifiers, splits are at match level, and the test set is the entire UEFA Women’s Euro 2025 tournament, disjoint from all training data.

¹ <https://github.com/xividi/vaep-phase-space>

3.5 Source-Target Training Strategies

We evaluate four strategies for combining the corpora: *women-only* training on the women’s 360 adaptation corpus alone, as a control; *men-only zero-shot* transfer; *fine-tuned*, pre-training on men’s data and continuing to boost on women’s data at a reduced learning rate; and *pooled* training on the union. Because pooled training also sees more data, a pooled advantage alone cannot distinguish cross-domain complementarity from a sample-size effect; we therefore add a size-matched comparison, subsampling the men’s corpus to the 95 matches of the women’s adaptation corpus, and learning curves computed separately per source.

3.6 Evaluation Metrics

Discrimination is measured by ROC-AUC and average precision (AP), the latter more sensitive to the rare positive class, given baseline positive rates of approximately 1.5% for scoring and 0.34% for conceding. Probability quality is measured by Brier score, log loss and Expected Calibration Error (ECE) [10]; because the conceding rate is so low that equal-width binning places nearly all actions in the lowest bin, ECE is reported over equal-frequency bins with a sensitivity check over the bin count. Player-valuation effects are reported through Spearman rank correlations.

Because PS-VAEP is defined only for actions with freeze-frame coverage (51,172 of 60,370 evaluation actions), every comparison must be matched, and all metrics here are reported on that identical subset. Reporting E-VAEP and P-VAEP on the full action set while restricting PS-VAEP to its own would compare different universes and systematically advantage PS-VAEP, since actions without a freeze frame are by definition harder broadcast-gap moments.

Uncertainty. All reported differences carry paired match-level bootstrap intervals [5]: we resample the 31 evaluation matches with replacement ($B = 1000$) and recompute every metric for every variant on each resample. Because actions within a match are correlated, the match rather than the action is the resampling unit, a cluster bootstrap [9], and since all variants share the evaluation action set, we form differences within each replicate and report percentile intervals for the difference. The player analysis uses the same unit, recomputing per-player minutes and VAEP per 90 within each replicate. The bootstrap captures evaluation-sample but not training uncertainty, so we additionally repeat every headline comparison across ten seeds; the two sources are of comparable magnitude, and reporting only one would substantially understate the uncertainty.

4 Results

4.1 Transfer Learning

Table 2 reports conceding-head ROC-AUC for the four strategies. Women-only training is weakest for every variant and pooled training best, and fine-tuning

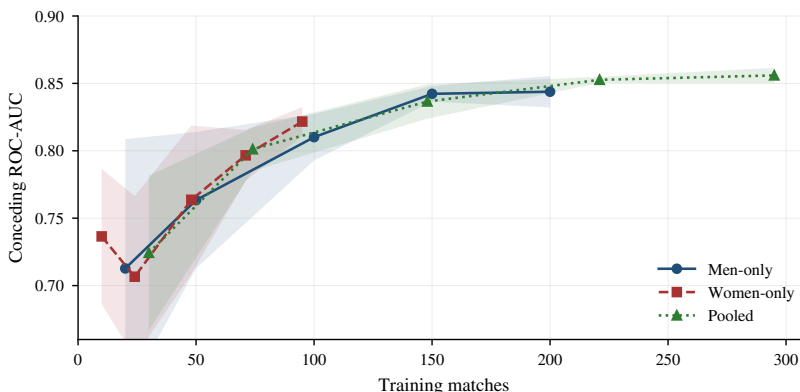


Fig. 1. Conceding ROC-AUC by training-set size and source composition; bands show ± 1 SD across seeds.

does not improve on zero-shot transfer, suggesting that continued boosting on the small target corpus adds less than joint training.

That ranking is confounded, however, because pooled training also sees more data. Fig. 1 therefore repeats the comparison at matched corpus size. At equal size the two single-domain strategies are indistinguishable (men-only 0.822 against women-only 0.821 over five seeds) and the three curves are close to superimposed: performance tracks the number of training matches, not their provenance. What the data support is therefore that additional 360 data helps regardless of provenance, not that cross-domain training carries complementary information — for a data-scarce target domain the more useful finding, since any available 360 corpus is then worth adding.

Table 2. Conceding ROC-AUC by training strategy on the matched 51,172-action subset (single seed).

Strategy	Matches	E-VAEP	P-VAEP	ES-VAEP	PS-VAEP
Women-only (control)	95	0.811	0.810	0.810	0.791
Men-only (zero-shot)	200	0.834	0.823	0.837	0.842
Fine-tuned	295	0.831	0.819	0.825	0.835
Pooled	295	0.837	0.865	0.840	0.849

The composition of the men’s source also matters, and not as gender alone would predict. Estimating a proxy $\mathcal{A}$ -distance with a domain classifier trained to separate action feature vectors, men’s international football sits at 0.713 from the women’s target while the men’s club corpus sits at 0.955, the two men’s subsets being only 0.214 apart. Size-matched transfer agrees: from 34 matches an international source reaches 0.755 and a club source 0.717. Competition type is therefore the more important axis of similarity here, and Bundesliga the more distant source despite matching the rest of the men’s corpus on gender.

4.2 Ablation: Event, Phase, and Space

Table 3 shows the matched ablation. Adding phase information improves the conceding head, and this is the only improvement anywhere in the ablation whose bootstrap interval excludes zero. The tabulated single run puts the paired effect at +0.027 (CI [+0.013, +0.046]), but seed variation is of the same order: averaged over ten seeds it is +0.006 (CI [+0.001, +0.011], positive in 8 of 10 seeds), with the tabulated run at the extreme of that distribution (Fig. 2a). We report the seed-averaged figure as the effect. The scoring head is unaffected, as Section 1 predicts.

Adding spatial features on top of phase does not produce a comparable gain; it produces a loss. PS-VAEP is below P-VAEP on both heads and on conceding AP (0.156 against 0.178), and the 2×2 design shows why this is not simply a failure of the spatial layer. The phase effect is +0.027 without space but +0.010 with it, while the space effect is +0.003 without phase and -0.015 with it, an interaction of -0.018 . The layers act as substitutes rather than complements: they encode overlapping information about how exposed a state is, and supplying both leaves the learner with noisier evidence than either alone. Neither main effect nor the interaction is individually separated from zero at $n = 31$ matches, so we present this as the pattern the design reveals rather than as three estimates. No log-loss or calibration difference is robust, so we make no calibration claim.

Table 3. Matched pooled-training ablation on 51,172 actions; bold denotes a paired bootstrap CI excluding zero.

	E-VAEP (event)	P-VAEP (+ phase)	ES-VAEP (+ space)	PS-VAEP (+ both)
Scoring ROC-AUC $\uparrow$	0.825	0.827	0.823	0.823
Scoring AP $\uparrow$	0.291	0.293	0.292	0.288
Scoring log loss $\downarrow$	0.0628	0.0626	0.0627	0.0629
Scoring ECE $\downarrow$ (10^{-3})	3.8	3.8	3.7	3.9
Conceding ROC-AUC	0.837	0.865	0.840	0.849
Conceding AP	0.171	0.178	0.167	0.156
Conceding log loss	0.0182	0.0178	0.0183	0.0182
Conceding ECE (10^{-3})	1.4	1.2	1.1	1.1

4.3 Where the Added Context Matters, and How Robust It Is

Stratified revaluation. To locate the effect of the spatial layer we decompose the per-action change $\Delta = V_{PS}(a) - V_P(a)$ across nearest-opponent distance, defensive density, pitch zone, action type and outcome (Fig. 2b). The result is negative and we report it as such. The stratum a congestion account predicts — three or more visible opponents within 5 m — shifts by +0.0010, interval $[-0.0003, +0.0023]$; nearest-opponent distance produces no gradient, and pitch zone none. The one robustly positive stratum is fouls (+0.0085), which no pressure account predicts. We therefore claim no mechanism: the spatial layer

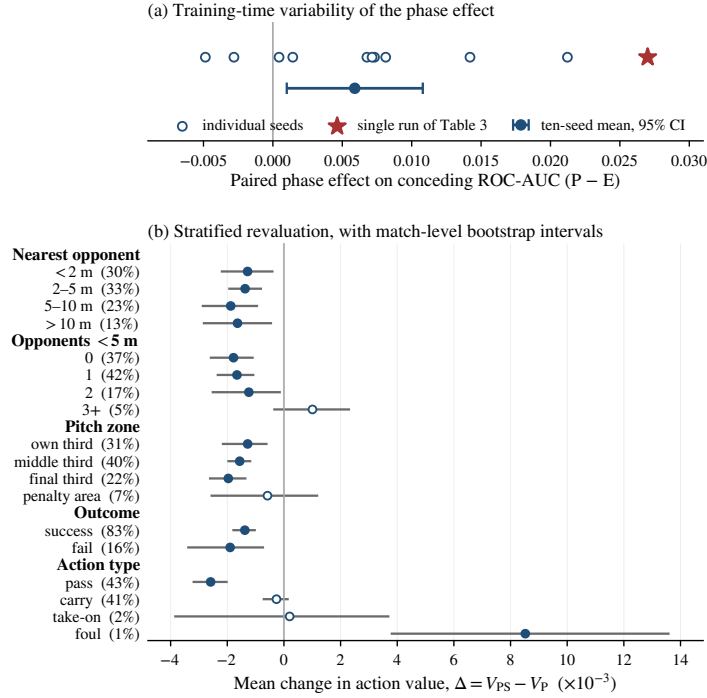


Fig. 2. Uncertainty analyses: (a) phase effect across ten seeds; (b) stratified revaluation $\Delta = V_{PS} - V_P$ with paired bootstrap intervals (filled: CI excludes zero).

demonstrably reallocates value, as Section 4.4 shows, but the stratification does not identify the situations responsible. An earlier version of this analysis, on models trained without the phase interactions, did find a robustly positive congestion stratum; that it did not survive retraining is itself a caution against reading mechanism into a redistribution.

Feature groups and event-data approximation. A leave-one-group-out analysis over the four feature groups finds only zone context to be load-bearing. The more consequential question, though, is whether the features are available at all without 360 coverage. Predicting each from event features alone, the zone indicators are recovered exactly — they are deterministic functions of the action coordinates — and the density and structure counts reach r^2 between 0.39 and 0.70. Rebuilt on these predictions, Pseudo-Space VAEP matches the observed-frame model and slightly exceeds it (0.858 against 0.853), so the spatial layer contributes little that event data does not already imply and the approach transfers to competitions without 360 coverage.

Sensitivity to modelling choices. Across ten seeds, conceding ROC-AUC has a standard deviation of 0.006 for E-VAEP and 0.009 for P-VAEP, and over a grid of 18 hyperparameter settings it ranges from 0.80 to 0.87 — wider than

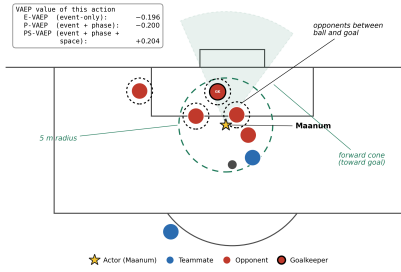


Fig. 3. Illustrative Maanum take-on with contrasting E/P-VAEP and PS-VAEP valuations.

any between-variant difference. Changing the learner preserves the ordering that matters (logistic regression 0.840 against 0.820; XGBoost [3] 0.865 against 0.860), so the phase advantage is a property of the feature set, not of the learner. Full tables accompany the code.

4.4 Player Valuation

While PS-VAEP does not improve aggregate prediction, it produces a materially different player ranking. On the 117 players with at least 270 minutes in the held-out tournament, aggregated over the identical matched action set, E-VAEP and P-VAEP produce almost the same ranking ($\rho = 0.977$), whereas PS-VAEP correlates with E-VAEP at $\rho = 0.824$ and with P-VAEP at $\rho = 0.808$. Match-level resampling shortens each player’s minutes and attenuates these correlations, so the intervals $([0.933, 0.976], [0.694, 0.845], [0.701, 0.841])$ are conservative and need not contain the point estimate. The spatial layer thus reorders the ranking to a degree the phase layer does not, even though it is phase that improves prediction. Individual rank changes are uncertain — only 25 of the 117 players have a change whose interval excludes zero — and the robust risers and fallers mix midfielders, forwards and defenders, so we do not claim the redistribution follows positional lines.

The matching requirement is consequential: aggregating E-VAEP over all 60,370 actions while PS-VAEP sees only 51,172 lowers the apparent correlation from 0.82 to 0.69 and inflates individual rank changes several-fold. We flag this because the artefact is easy to introduce and flatters the spatial model.

We deliberately do not frame this redistribution as a correction. Reduced credit for progression through open space need not indicate over-valuation by the event-only model: such progression may be legitimately valuable, and the space it exploits may have been created by preceding off-ball movement. Space creation is invisible to an on-ball model, so the downward shift of deep-lying progressors is as consistent with a blind spot in PS-VAEP as with an error in E-VAEP.

Case study: a take-on under pressure. Fig. 3 shows one action-level case. In a take-on by Frida Maanum against Iceland, centrally in the penalty area with four

opponents within 5 m and four between ball and goal, E-VAEP and P-VAEP assign -0.196 and -0.200 whereas PS-VAEP assigns $+0.204$, consistent with the model treating ball retention under dense pressure as valuable. This illustrates how spatial features enter valuation; it is not evidence for a mechanism.

5 Discussion and Limitations

The results separate two effects in action valuation. Phase context improves conceding-head probability estimation, the only effect separated from zero under both match-level resampling and seed variation. Spatial context does not improve aggregate prediction and degrades it when added to phase, but substantially changes player valuation, suggesting different roles on the predictive and allocation sides.

The stratified analysis does not identify which situations drive this reallocation, and an earlier pattern did not survive retraining. Thus, contextual valuation should not be judged only by ROC-AUC or log loss: similar predictive performance can coexist with different allocations of credit. PS-VAEP is therefore better interpreted as a revaluation model than as a predictively superior one; whether its redistribution is preferable requires independent expert validation.

Two further findings emerge. Source similarity depends strongly on competition type, while at matched corpus size men’s and women’s data provide comparable predictive value, suggesting that data volume matters more than provenance. Moreover, spatial features predicted from event data closely reproduce the observed-frame model, which is particularly relevant in women’s football, where event data are much more widely available than spatial data.

The main limitations are evaluation on a single held-out tournament, partial freeze-frame visibility, on-ball-only valuation, and a rule-based phase taxonomy. Although coverage varies by zone (14.4 visible players in the middle third versus 12.2 in the penalty area), its association with revaluation is negligible (Spearman 0.020 overall, 0.065 within the penalty area). Finally, effect sizes are comparable to training-time variability, motivating seed-averaged reporting. Overall, different context layers affect different aspects of valuation rather than uniformly improving prediction.

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